ELEC2134 Laboratory 3

Second-Order Responses

Overview

During this lab you will investigate the natural and transient responses of RLC circuits. The transient response will be investigated by varying the circuit resistance with your potentiometer, allowing you to watch the response change fluidly from overdamped, to underdamped.

Aims

- Investigate the transient response of a second order parallel RLC circuit.
- Observe and determine the conditions for different responses in second-order RLC circuit.

Theory

Consider the RLC circuit shown in Figure 3.1. Assume initial inductor current I_0 and initial capacitor voltage V_0.

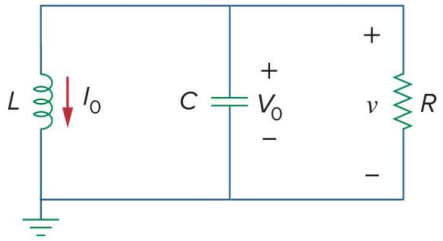


Figure 3.1. Natural response of parallel RLC circuit.

The analysis of this circuit yields the following second-order differential equation.

$$\frac{d^2v}{dt^2}+\frac{1}{RC}\frac{dv}{dt}+\frac{v}{LC}=0$$

The characteristic equation of this differential equation can be written as $s^2+\frac{1}{RC}s+\frac{1}{LC}=0$, which has the following roots:

$$s_{1,2}=-\alpha\pm\sqrt{\alpha^2-\omega_0^2},$$

where the neper frequency (Np/s) is $\alpha=\frac{1}{2RC}$, and the resonant frequency (rad/s) is given by $\omega_0=\frac{1}{\sqrt{LC}}$.

These values determine the natural and step response of the circuit.

There are three possible solutions:

- **Overdamped ($\alpha^2>\omega_0^2$)**

  The response is the sum of two decaying exponentials. There is no oscillation in the response.

- **Underdamped ($\alpha^2<\omega_0^2$)**

  The response is the sum of two sinusoids multiplied by a decaying exponential. The response oscillates around the final value.

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- **Critically damped ($\alpha^2 = \omega_0^2$)**

  This is the border case between the overdamped and underdamped responses. It is the fastest the response can reach its final value without oscillation.

From this, we can see that by appropriate selection of components we can produce all the types of damping from the response to the parallel RLC circuit.

Prelab Preparation

An automobile airbag igniter is modelled by the circuit in Figure 3.2, where the igniter is powered by the car battery and modelled as a resistor R. Let $R = 1000\,\Omega$, $C = 100\,\text{nF}$ and $L = 10\,\text{mH}$.

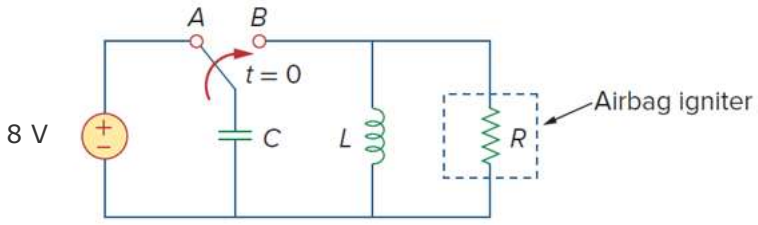


Figure 3.2. Automobile airbag igniter circuit.

1. Derive a differential equation to describe the voltage across the igniter and solve it in the time domain.
2. Verify your answers using LTSpice.
   Hint: Due to some limitation of switches in LTSpice, the simplest way to verify your answer is to draw two schematics. One is for the circuit when $t < 0$, where you conduct transient simulation to obtain the initial condition of capacitor C. Then you draw a schematic for the circuit when $t > 0$ and let LTSpice know the initial conditions when you do the simulation to observe $v(t)$. To add initial conditions, go to 'SPICE Directive' from 'Edit' drop-down menu. Type in '.ic V(n001)=a I(L1)=b', where *a* is the initial voltage of capacitor C and b is the initial current through the inductor L. In the simulation command, you need to tick 'Start external DC supply voltages at 0V' to avoid 'singular matrix' error.
3. Use the simulation in question 2 to explore the effect of each of the circuit elements (R, C and L) in the value of the peak time and other properties of the transient response.

   iv. Change the value of R and discuss its effect.

   v. Change the value of C and discuss its effect.

   vi. Change the value of L and discuss its effect.

   vii. Based on your observations, suggest a way to increase the amplitude of the first positive peak voltage.

Once you have completed all the questions from this checkpoint, please submit a marking request here: https://forms.cloud.microsoft/r/9mnwX47BP1.

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Lab Exercises

Section 1. Natural Response

The natural response of the parallel RLC circuit in Figure 3.1 is found by calculating the voltage v across the parallel components due to the energy in the inductor and/or capacitor as a result of their initial conditions I_0 and V_0. This is equivalent to the circuit in Figure 3.3 below, which you analysed in the Prelab for this laboratory.

When the switch is in position A, the capacitor has a voltage equal to the voltage supplied from the battery. When it is in position B, the battery is disconnected and the capacitor discharges through the parallel elements.

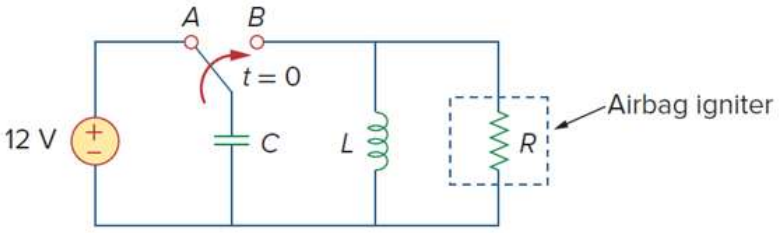


Figure 3.3. Automobile airbag igniter circuit.

To experimentally observe the natural response, you will use the test circuit in Figure 3.4. The voltage source is a positive square wave with amplitude V_s from 0V to 8V and period T. This allows the natural response to be generated periodically while V_s stays at 0V, provided that T is long enough to allow the energy-storing elements (capacitor and inductor) to reach steady-state as they charge and discharge.

In the circuit, R_s represents the internal resistance of the voltage source (50Ω). Therefore, the circuit in Figure 3.4 is equivalent to the one in Figure 3.3 with the switch in position B when i) $V_s = 0V$ and ii) $R = R_s + R_c$ (note that while the latter would make both circuits completely equivalent, the resistance of the airbag igniter cannot change, so we will still use $R_c = 1000\Omega$ in the experiment to represent the airbag igniter, as in the Prelab).

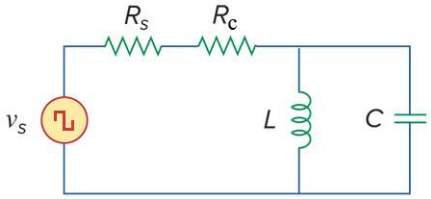


Figure 3.4. Parallel RLC circuit test circuit.

Preparation

Using $L = 10\text{mH}$ and $R_c = 1000\Omega$, determine the value of C that will result in a critically damped response. To obtain an underdamped response, would C need to be less than, or greater than this value?

Experiment

Construct the circuit in Figure 3.4 using $L = 10\text{mH}$, $R_c = 1000\Omega$ and $C = 100\text{nF}$.

1. Using the signal generator apply an 8V peak-to-peak square wave with 4V offset such that the negative pulse voltage is at 0V. Set the frequency to $f \approx 100\text{Hz}$ or less so that the circuit can return to its steady state value in each half period of the square wave.

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Note: Please ensure that Channel 1 is DC coupled. To do this, push the "1" button under the vertical scale knob. This will display the menu for coupling. The coupling menu determines if the DC component of the signal will be displayed on the screen or not.

2. Measure the voltage from the signal generator using CH1 and the voltage across the capacitor using CH2. You should see the natural response of the circuit. Is this similar to your solution to Question 1 in your Prelab? Sketch the waveform in the oscilloscope display in Figure 3.5.

Note: You may need to increase the horizontal scale to observe the transient response.

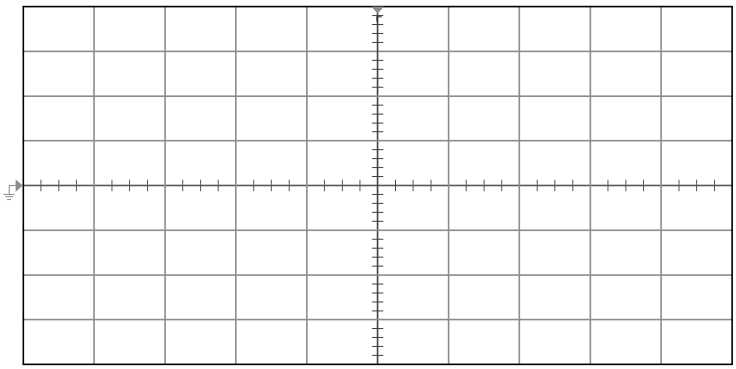


Figure 3.5. Display of an oscilloscope.

3. Now replace the capacitor with the value you calculated for the critically damped response. Does this agree with your preparation? Sketch the waveform in the oscilloscope display in Figure 3.6.

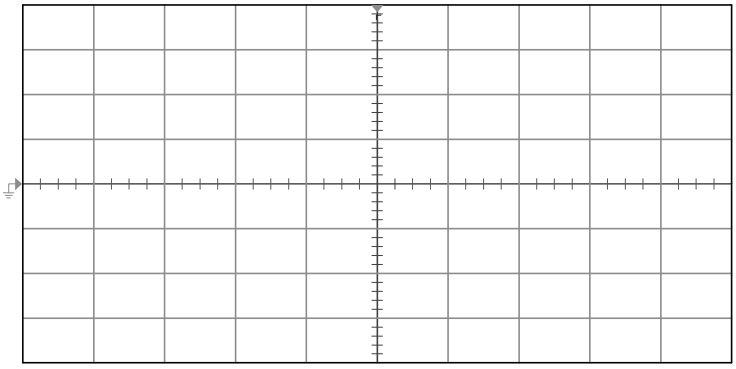


Figure 3.6. Display of an oscilloscope.

4. Can you create an overdamped response by using a different capacitor value? Sketch the waveform in the oscilloscope display in Figure 3.7 and compare the three different cases.

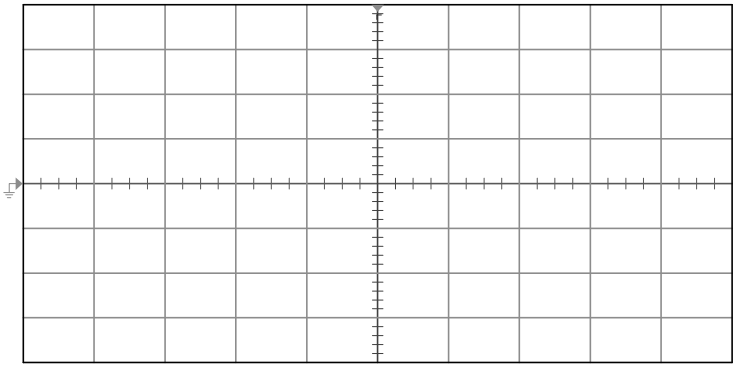


Figure 3.7. Display of an oscilloscope.

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Section 2. Step Response

The step response of the parallel RLC circuit can be observed by measuring the current i through the inductor when we apply a sudden constant current I_s, as shown in Figure 3.8. In this case we want to measure the current i through the inductor as, unlike what happens with the resistor and capacitor, the current does not go to zero as $t \to \infty$. When testing the step response, we are interested in the forced response of the circuit.

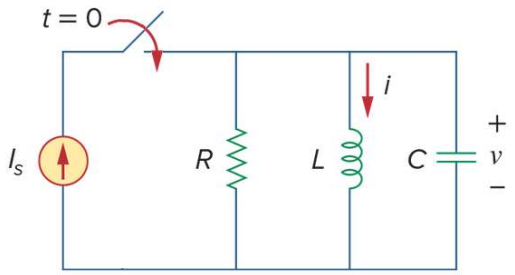


Figure 3.8. Parallel RLC circuit step response

As we do not have access to a constant current source in the laboratory, we instead perform a source transformation. The current source I_s in parallel with the resistor R is transformed into the voltage source $V_s = I_s R$ in series with R. We break this resistance into the sum of the source's internal resistance R_s and our component selection R_c to get our test circuit, as shown in Figure 3.9. As the oscilloscopes cannot measure current, we add a small test resistor R_{test} of value 10Ω in series with the inductor to measure a voltage proportional to the inductor's current $i = \frac{v_o}{10}$.

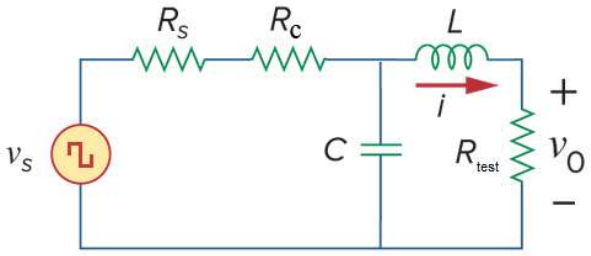


Figure 3.9. Parallel RLC test circuit to observe step response.

Experiment

Use $L = 10\text{mH}$ and $C = 10\text{nF}$. To dynamically adjust the resistance, we will be using a **$10k\Omega$ potentiometer** for R_c.

1. Use the digital multimeter to adjust the potentiometer such that the resistance between the **black and yellow leads** is equal to 1000Ω.

2. Modify the circuit that you set up for Section 1 to match the circuit in Figure 3.9 by adding the 10Ω test resistor, replacing the capacitor with one of value 10nF[2], and replacing the 1000Ω resistor with the potentiometer, using the black and yellow leads. Please leave the potentiometer's red lead unconnected. Use the same input square wave as in Section 1.

3. Measure input voltage source using CH1 and the voltage v_0 across the test resistor using CH2.

4. Using the cursors, measure the rise time and record this value in Table 3.1, and sketch the waveform in Figure 3.10.

   Note: Rise Time is the time required for a signal or electronic waveform to transition from a low-voltage level to a high-voltage level, typically measured between 10% and 90% of its final amplitude.

5. Adjust the potentiometer's resistance until the response becomes critically damped (the first point it ceases oscillating). Again, sketch the waveform in Figure 3.10, record the resistor value and rise time in Table 3.1.

6. Create an overdamped response by increasing or decreasing the value of the potentiometer. Sketch the waveform in Figure 3.10, record the resistor value and rise time in Table 3.1. Explain why this new resistance value caused the circuit to be overdamped.

7. In control systems applications, it is sometimes desirable to achieve a 20% overshoot step response. By varying circuit components, can you produce such a response? Record the component values you use, and sketch the response in Figure 3.11.

Table 3.1. Response performance of a second-order system under different damping conditions

	Resistance from Potentiometer, R_c	**Rise Time**
Underdamped		
Critically-damped		
Overdamped		

[2] Note that this new capacitance value provides much more visible results, allowing you to clearly observe the transitions between overdamped, critically damped and underdamped.

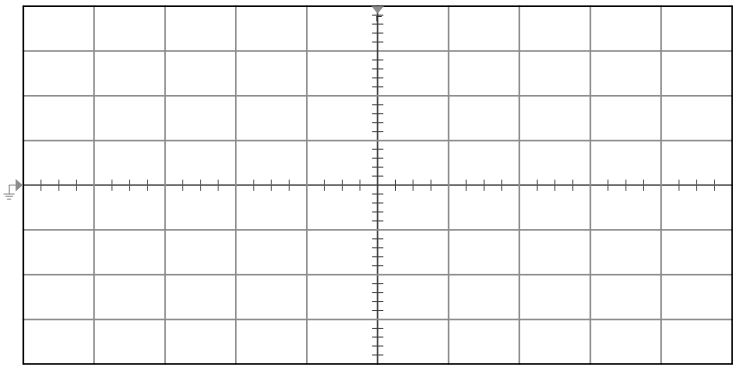


Figure 3.10. Display of an oscilloscope.

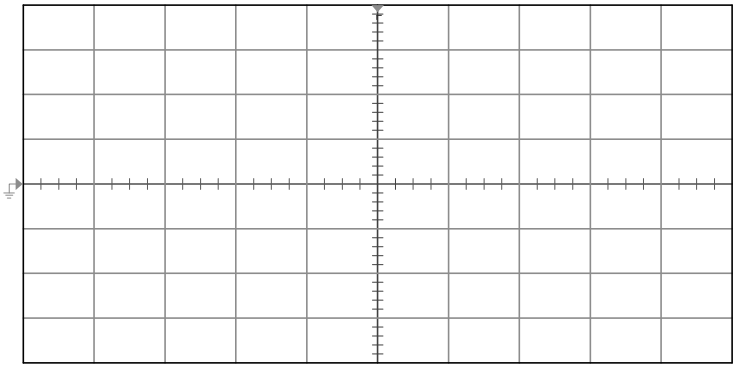


Figure 3.11. Display of an oscilloscope.

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